Forensic Statistics Audit

Rajizadeh et al., "Effect of magnesium supplementation on depression status,"
Nutrition 35 (2017) 56–60

Generated by Paperscope
`https://github.com/todd866/paperscope`

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What This Report Is

This report checks whether the **numbers** in a published scientific paper are internally consistent. It does not evaluate whether the study was well-designed or whether magnesium actually helps depression. Instead, it asks a simpler question: *could the reported statistics have come from real data?*

The answer, for this paper, is no — not for all of them.

We used 19 automated checks based on techniques from the forensic metascience literature, primarily Heathers (2025).[1] These same techniques have been used to identify data problems in hundreds of published papers, leading to corrections and retractions.

What the Paper Claims

Rajizadeh et al. conducted a small trial with 60 depressed patients who also had low magnesium. Half received magnesium supplements; half received a placebo. After 8 weeks, they measured depression scores and blood magnesium levels.

The paper concludes that magnesium supplementation improves both depression and magnesium levels. The trial was published in *Nutrition*, a peer-reviewed Elsevier journal.

What We Found: The Summary

We ran 50 individual checks across the paper's three data tables. The results:

Category	Count	Meaning
Failures	**14**	Mathematically impossible
Flags	**7**	Suspicious but not proven impossible
Passes	29	Checks out fine
Total	**50**	

Verdict: **severe data integrity concerns**. Multiple reported statistics cannot have come from the data described.

[1] Heathers, J. (2025). *An Introduction to Forensic Metascience*. DOI: 10.5281/zenodo.14871843

1 The Checks, Explained

1.1 GRIM: Can This Average Be Real?

The idea: The Beck Depression Inventory (BDI) is a questionnaire where patients score each of 21 items as 0, 1, 2, or 3. The total is always a whole number between 0 and 63. When you average whole numbers from a known number of people, only certain averages are mathematically possible.

Example: If 26 people each score a whole number, the group average must be some whole number divided by 26. The possible averages near 11 are $284/26 = 10.923$, $285/26 = 10.961$, $286/26 = 11.000$, $287/26 = 11.038$, and so on. The reported average of 11.26 would require $26 \times 11.26 = 292.76$ — but you can't have 0.76 of a point. No combination of 26 whole numbers produces an average of exactly 11.26.

Result:

Group	Timepoint	Mean	Verdict
Magnesium	Baseline	26.9	Pass
Magnesium	End	11.26	**Impossible**
Placebo	Baseline	25.6	Pass
Placebo	End	15.2	Pass

1.2 SPRITE: Can Any Dataset Produce These Statistics?

The idea: GRIM checks the average alone. SPRITE goes further: given the reported average *and* the spread (standard deviation), is there *any* possible set of whole-number BDI scores that produces both? The algorithm randomly shuffles candidate datasets and tries to match both statistics simultaneously.

Result: Using 50,000 iterations per seed with 3 independent seeds per target, three of the four BDI mean/SD pairs could not be reconstructed. One (placebo end) was found in 515 attempts, confirming it is achievable. A separate deep search using 2 million iterations per seed confirmed the same pattern, with the closest near-misses still falling short.

1.3 Correlation Bound: Are the Spreads Mutually Consistent?

The idea: When you measure something before and after treatment, you get three spreads: the spread of "before" scores, the spread of "after" scores, and the spread of the *changes*. These three numbers are mathematically linked by a formula involving the correlation between before and after. If you plug in the reported numbers and the formula says the correlation must be greater than 1 (or less than -1), the numbers are impossible — correlations are bounded between -1 and $+1$.

Result: For the placebo group's serum magnesium measurements:

- Before SD = 0.13, After SD = 0.27, Change SD = 0.03
- Implied correlation: $r = 1.27$ — **impossible**.
- The smallest possible change SD for these before/after SDs is 0.14, not 0.03.

This is one of the strongest findings. The reported change SD of 0.03 is off by nearly a factor of 5 from the minimum that mathematics allows.

1.4 P-Value Recalculation

The idea: A p-value is calculated from the average, the spread, and the sample size using a specific formula. We can redo the calculation and check whether the reported p-value matches.

Result: The *between-group* comparisons (the main results) check out — the reported p-values match our calculations. But the *within-group* paired tests are off by orders of magnitude:

Test	Reported p	Calculated p	Verdict
Placebo serum Mg change	0.110	1.0×10^{-14}	**Flag**
Mg group serum Mg change	0.001	1.2×10^{-5}	**Flag**
Mg group BDI change	0.001	2.9×10^{-9}	**Flag**
Placebo BDI change	0.001	2.9×10^{-7}	**Flag**

The placebo serum Mg result is the most striking: the reported p = 0.110 (not significant) should be $p \approx 10^{-14}$ (astronomically significant) given the reported numbers. This is likely related to the impossible change SD of 0.03 discussed above.

1.5 Arithmetic Consistency: Do the Tables Add Up?

The idea: If baseline is 63.33 and endpoint is 71.51, then the change should be $71.51 - 63.33 = 8.18$. We check every row.

Result: Six of eight dietary intake rows in Table 2 fail this check, with discrepancies ranging from 28 to 169 units.

One hypothesis (proposed by Meyerowitz-Katz on PubPeer) is that the mean and standard deviation columns were accidentally swapped. This is plausible because, for example, in the placebo protein row the actual change is $69.67 - 75.08 = -5.41$, and the number reported in the "SD" position is -5.40 — matching the true change, not a spread measure. The number in the "mean change" position (23.45) would then be the actual SD.

Additionally, three reported standard deviations in Table 2 are *negative* (-5.40, -55.89, -5.56). Standard deviations measure spread and cannot be negative. If these are actually mean changes that ended up in the wrong column, the sign makes sense. However, we note this is an interpretation of the pattern, not a proven fact — other explanations (e.g., data entry errors from a different source) are possible.

1.6 Carlisle Test: Was the Randomization Real?

The idea: In a properly randomized trial, the baseline characteristics of the two groups should differ by chance. The p-values comparing groups at baseline should be uniformly distributed. If they are *too* perfectly balanced, the randomization may have been manipulated.

Result: The Table 1 baseline p-values (0.93, 0.28, 0.80, 0.67, 0.89) pass the Carlisle test — they are consistent with real randomization. This is actually reassuring: the data integrity problems appear to be in the *outcome* reporting, not the trial design.

1.7 Other Checks

- **GRIMMER** (SD consistency for integer data): the Mg group endpoint fails because the mean already fails GRIM.
- **Chi-squared recalculation** on Table 1: sex and marital status p-values match.
- **SD/SE confusion**: the placebo serum Mg change SD of 0.03 is flagged as possibly being a standard *error* rather than a standard deviation — if it were the SE, the true SD would be $\sim$0.16, which is within plausible range.

- **Effect size consistency**: the between-group BDI effect (Cohen's $d = -0.62$, $p = 0.028$) is internally consistent.
- **Variance ratios**: group variances are not suspiciously similar or different.
- **Quick SD check**: all BDI standard deviations are plausible for the 0–63 range.

2 What Does This Mean?

The pattern is clear:

1. **Table 1** (baseline characteristics) and **between-group comparisons** are internally consistent. The randomization looks real, and the main treatment effect statistics check out.
2. **Table 2** (dietary intake changes) has systematic errors consistent with a mean/SD column swap during manuscript preparation.
3. **Table 3** (outcomes) contains at least one impossible mean, an impossible correlation bound, and p-values that don't match the reported statistics.

These findings do **not** prove fraud. Possible explanations include transcription errors, software misuse, rounding cascades, or subset confusion (e.g., running an analysis on 25 people but reporting $n = 26$). However, they make it impossible to evaluate the paper's claims from the published numbers alone.

The appropriate response, as recommended by forensic metascience practice, is to request the raw dataset from the authors so these discrepancies can be resolved.

3 How This Analysis Was Produced

This report was generated using **Paperscope**, an open-source toolkit for academic paper analysis. The forensic statistics module implements 19 checks based on techniques described in Heathers (2025), Brown & Heathers (2017), and related work.

Important methodological note: The values tested in this audit were *manually transcribed* from the published tables in Rajizadeh et al. (2017), not automatically extracted from the PDF. This paper serves as the built-in demonstration case for the forensic module. The checks themselves are automated — once the numbers are entered, no subjective judgment affects the results — but the initial data entry step is manual and therefore subject to transcription error. All values can be verified against the original paper (Tables 1–3, pages 58–59).

In a typical review workflow, the reviewer reads the paper, transcribes the relevant summary statistics, and feeds them to the forensic functions. Automated PDF table extraction is a separate capability not used here.

Software: `https://github.com/todd866/paperscope`
Module: `paperscope.analysis.forensic_stats`
Reference: Heathers, J. (2025). *An Introduction to Forensic Metascience.* DOI: 10.5281/zenodo.14871843

A Provenance Table

Every tested value is listed below with its source location in the original paper, the check applied, and the result.

Source	Value	Check	Expected	Result
Table 3, Beck score means (integer instrument, BDI range 0–63)				
T3, Mg baseline	mean=26.9, n=26	GRIM	n×mean $\in Z$	Pass
T3, Mg end	mean=11.26, n=26	GRIM	n×mean $\in Z$	Fail: 292.76
T3, Placebo baseline	mean=25.6, n=27	GRIM	n×mean $\in Z$	Pass
T3, Placebo end	mean=15.2, n=27	GRIM	n×mean $\in Z$	Pass
Table 3, Beck score mean+SD pairs (SPRITE, 50k iter × 3 seeds)				
T3, Mg baseline	26.9 ± 7.1, n=26	SPRITE	any valid dataset	Not found
T3, Mg end	11.26 ± 6.9, n=26	SPRITE	any valid dataset	Fail (GRIM)
T3, Placebo baseline	25.6 ± 6.1, n=27	SPRITE	any valid dataset	Not found
T3, Placebo end	15.2 ± 9.3, n=27	SPRITE	any valid dataset	Found (iter 515)
Table 3, Serum Mg pre/post/change SDs (correlation bound)				
T3, Placebo Mg	SD: 0.13, 0.27, 0.03	Corr. bound	$\lvert r\rvert \leq 1$	Fail: r=1.27
T3, Mg group	SD: 0.19, 0.19, 0.29	Corr. bound	$\lvert r\rvert \leq 1$	Pass: r=−0.16
Table 3, Paired t-test p-values				
T3, Placebo Mg change	p=0.110	Recalc	match	$p_{calc}=10^{-14}$
T3, Mg group Mg change	p=0.001	Recalc	match	$p_{calc}=10^{-5}$
T3, Mg BDI change	p=0.001	Recalc	match	$p_{calc}=10^{-9}$
T3, Placebo BDI change	p=0.001	Recalc	match	$p_{calc}=10^{-7}$
Table 3, Between-group t-tests				
T3, BDI baseline	p=0.49	Recalc	match	Pass: 0.48
T3, BDI end	p=0.08	Recalc	match	Pass: 0.09
T3, BDI change	p=0.02	Recalc	match	Pass: 0.03
T3, Mg baseline	p=0.27	Recalc	match	Pass: 0.27
Table 2, Change scores (End − Baseline = Change?)				
T2, Protein, Mg	change=44.31	Arithmetic	8.18	Fail (Δ36)
T2, Protein, Placebo	change=23.45	Arithmetic	−5.41	Fail (Δ29)
T2, Carbs, Mg	change=87.21	Arithmetic	2.39	Fail (Δ85)
T2, Carbs, Placebo	change=113.34	Arithmetic	−55.89	Fail (Δ169)
T2, Fat, Mg	change=−6.35	Arithmetic	−6.35	Pass

Source	Value	Check	Expected	Result
T2, Fat, Placebo	change=32.15	Arithmetic	-5.57	Fail (Δ38)
T2, Diet. Mg, Mg	change=81.50	Arithmetic	17.36	Fail (Δ64)
T2, Diet. Mg, Placebo	change=90.29	Arithmetic	30.14	Fail (Δ60)
Table 2, Negative SDs				
T2, Protein Δ, Placebo	SD=-5.40	SD ≥ 0	non-negative	Fail
T2, Carbs Δ, Placebo	SD=-55.89	SD ≥ 0	non-negative	Fail
T2, Fat Δ, Placebo	SD=-5.56	SD ≥ 0	non-negative	Fail
Table 1, Baseline p-values (Carlisle test + chi-squared recalc)				
T1, 5 baseline p-values	0.93–0.89	Carlisle	uniform on [0,1]	Pass: Z=-1.52
T1, Sex	χ^2, p=0.93	Chi-sq recalc	match	Pass: p=0.93
T1, Marital status	χ^2, p=0.28	Chi-sq recalc	match	Pass: p=0.29